

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

Data Scientist - Fedway Associates, EliteUS Software Solutions Aug 2024 - Present | Edison, NJ

- Designed and deployed predictive ML models on 15M+ interaction records, improving user engagement by 30% and translating product requirements into measurable business impact.
- Built classification and regression models for user behavior prediction and engagement scoring using feature engineering and offline/online validation to optimize business KPIs.
- Performed large-scale behavioral analysis using SQL/Python to identify usage patterns and generate insights for product teams; developed LLM-based search systems using RAG for enterprise knowledge retrieval.
- Implemented end-to-end LLM evaluation frameworks (golden sets, automated metrics, A/B testing, human review loops) improving response quality by 30% through prompt and model iteration.

Data Science Intern, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida

- Engineered scalable ML pipelines and monitoring systems for large advertising datasets, increasing campaign visibility from 15% to 45% while ensuring reliability and performance.
- Built ML services and NLP APIs using XGBoost, H2O AutoML, and FastAPI, improving sentiment classification accuracy by 65% and campaign spend efficiency by 50%.
- Designed SQL/Python pipelines for feature engineering, dataset validation, and model training, enabling repeatable experimentation, evaluation, and production handoff of ML models.

Application Development Associate, Accenture June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, and indexed lookups to efficiently clean, deduplicate, transform large datasets, reducing 50% production issues.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations to data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming: Python, R, C++, SQL. **Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
Machine Learning: End-to-end ML pipelines, model training, evaluation, monitoring, feature engineering, MLOps.
GenAI & LLMs: Transformer-based models, LLM fine-tuning, retrieval-augmented generation, multimodal modeling.
Evaluation: Golden datasets, human-in-the-loop review, automated metrics, offline/online evaluation frameworks.
Cloud & MLOps: PostgreSQL, Data Warehousing, AWS (Sagemaker, EMR), GCP, Azure, Pinecone, Chroma.
Statistics: A/B & hypothesis testing, statistical inference, experiment design, regression analysis, causal inference.

EDUCATION

Rutgers University, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ

Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

LLM-Powered Retrieval & Question Answering System (Production): Built and shipped LLM-powered retrieval and question answering services using embeddings, RAG, and evaluation-driven iteration, improving response quality, factual grounding, and production reliability.

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>